Notes on the GUNDAM fit for the ICARUS NuMI ν_eCC inclusive analysis

Dial interpolation, fit procedure, errors, and likelihood scans

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Contents

1 Scope and setup **1**

2 How the GUNDAM fit is performed **1**
2.1 The model prediction . . . 1
2.2 The likelihood . . . 2
2.3 Minimization . . . 2

3 GENIE dials: splines, graphs and morph knobs **2**
3.1 Two kinds of GENIE knob . . . 2
3.2 What is actually stored . . . 3
3.3 Spline vs Graph dials . . . 3
3.4 Mirroring . . . 3
3.5 Options and recommendation . . . 4

4 Errors and the Hessian **5**
4.1 From curvature to covariance . . . 5
4.2 What contributes to the curvature . . . 5
4.3 Marginal vs conditional errors . . . 5
4.4 When the Hesse error can be trusted . . . 5
4.5 Bounded parameters: the morph knobs on $[0, 1]$ . . . 6

5 Likelihood scans **7**
5.1 What the scan produces . . . 7
5.2 Reading the shapes . . . 7
5.3 Using scans to decide whether the fit succeeded . . . 8
5.4 Unconstrained systematics are not bad systematics . . . 8

6 Why the extracted cross-section uncertainty is large **8**
6.1 The flux–template degeneracy . . . 9
6.2 Why the distribution is precise but the cross section is not . . . 9
6.3 Pre-fit vs post-fit statistical band: why the post-fit is larger . . . 10
6.4 Decomposition of the total uncertainty . . . 11
6.5 How each contribution is computed (`--decompose`) . . . 12

6.6 The diagonal-response (perfect-resolution) test 13
6.7 More statistics does not help 14
6.8 Summary: three mechanisms that inflate the cross section 14
6.9 What would reduce it 15

7 Checklist for a fit **15**

1 Scope and setup

These notes collect what was learned while debugging the GENIE morph knobs in the fake-data studies, and explain how the GUNDAM fit works well enough to judge whether a given fit can be trusted. Unless stated otherwise, everything refers to **GUNDAM 1.9.X** as built in `GUNDAM-1.9.X/`, and to the configs in `configsNue/`.

Item	Setting
Minimizer	Minuit2, Migrad (Simplex first), strategy 1, tolerance 10^{-2}
Errors	Hesse (`errors: "Hesse"`)
Fit space	normalized (`useNormalizedFitSpace: true`)
Likelihood	`PoissonLLH`
Parameter sets	template (true E_e bins), GENIE, NuMI beamline, detector
Input MC	`gundam_v30.root`, tree `events/NuMIFull/selected`

The minimizer config is `Configs_Minimizer/default_minimizer.yaml`; the GENIE parameter set is `Configs_ParameterSet/GENIE/`.

2 How the GUNDAM fit is performed

2.1 The model prediction

Every MC event e that passes the global and sample selection cuts carries a nominal weight w_e^{nom}, built from `nominalWeightFormula` (POT scaling × `ppfx_cv_weight` for the neutrino MC). Each parameter θ_k that applies to the event contributes a *dial response* $r_{ek}(\theta_k)$, so the event weight is

$$w_e(\boldsymbol{\theta}) = w_e^{\text{nom}} \prod_k r_{ek}(\theta_k). \tag{1}$$

For the GENIE knobs, r_{ek} is interpolated per event from the `TClonesArray` of TGraphs stored in the input tree (Section 3). For the template parameters, $r_{ek} = \theta_k$ for events in true-E_e bin k, and 1 otherwise.

The predicted content of reconstructed bin b is then $\mu_b(\boldsymbol{\theta}) = \sum_{e \in b} w_e(\boldsymbol{\theta})$, and d_b is the (fake) data in the same bin.

Per-dataset selection is possible: each `mc` and `data` block in a dataset accepts `selectionCutFormula` (a string, or a list joined with `&&`), applied before the per-sample cuts.

2.2 The likelihood

GUNDAM minimizes a quantity on the $-2\ln L$ (i.e. χ^2-like) scale:

$$\mathcal{L}(\boldsymbol{\theta}) \equiv \text{llh} = \underbrace{\sum_b 2\left[\mu_b - d_b + d_b \ln \frac{d_b}{\mu_b}\right]}_{\text{llhStat}} + \underbrace{\sum_{\text{sets}} \boldsymbol{\Delta}^T V^{-1} \boldsymbol{\Delta}}_{\text{llhPenalty}}, \qquad \boldsymbol{\Delta} = \boldsymbol{\theta} - \boldsymbol{\theta}^{\text{prior}}. \tag{2}$$

The stat term is the Baker–Cousins Poisson form (`PoissonLogLikelihood.h`); bins with $d_b = 0$ contribute $2\mu_b$, and a bin with $\mu_b \le 0$ gives $+\infty$. The penalty term is computed per parameter set from its prior covariance V (`LikelihoodInterface.cpp`); a set with no covariance matrix, such as the template parameters, contributes nothing and is constrained only by the data.

Because `llh` is on the $-2\ln L$ scale, Minuit's default `ErrorDef` $= 1$ is the correct one: $\Delta\text{llh} = 1$ defines the 1σ interval for a single parameter.

2.3 Minimization

1. **Normalized fit space.** With `useNormalizedFitSpace: true` the minimizer sees $\tilde{\theta}_k = (\theta_k - \theta_k^{\text{prior}})/\sigma_k$, so every parameter has prior 0 and width 1. Covariances are converted back afterwards by multiplying by $\sigma_i \sigma_j$ (`RootMinimizer.cpp`).
2. **Simplex** runs first with a loose tolerance to get near the minimum.
3. **Migrad** then descends using the gradient and an iteratively built approximation to the Hessian, stopping when the estimated distance to the minimum (EDM) is below $0.002 \times$ tolerance (Minuit2 convention, UP = 1).
4. **Hesse** recomputes the full matrix of second derivatives at the minimum by finite differences and inverts it to get the post-fit covariance (Section 4).

Parameters with bounds (`parameterLimits`, or the lower/upper-bound vectors in a covariance file) are handled by Minuit through a sine transform to an unbounded internal variable. This matters for errors near a bound (Section 4.4).

The post-fit values are written to `FitterEngine/postFit/parState_TNamed` and the covariance to `FitterEngine/postFit/Hesse/hessian/`; the toy generator and `gundamCalcXsec` read these back.

3 GENIE dials: splines, graphs and morph knobs

3.1 Two kinds of GENIE knob

Multisigma knobs (e.g. `MaCCRES`, `NormCCMEC`) scale a model parameter by some number of standard deviations. The response is meaningful for both signs, and $\theta = \pm 1$ is a genuine $\pm 1\sigma$ variation.

Morph knobs (`VecFFCCQEshape`, `DecayAngMEC`, `Theta_Delta2Npi`, `ThetaDelta2NRad`) interpolate between two models: $\theta = 0$ is the GENIE nominal and $\theta = 1$ is the alternative model. Negative values have no model behind them.

In the covariance file built by `GENIE/makeParameterRootFile.C`, both kinds get prior 0, $\sigma = 1$, and no correlations (only the four Z-expansion knobs are correlated). The one exception is a prior of 0.005 on `VecFFCCQEshape`, which appears to be a workaround for the zero-slope problem described below.

3.2 What is actually stored

Inspecting the first 11,729 events of `gundam_v30.root` shows that all four morph knobs are stored at $x = \{-1, -0.5, 0, 0.5, 1\}$ and are **exactly symmetric**, $w(-x) = w(x)$, in every event with a non-trivial response:

Knob	Non-flat events	Example $w(x)$ at $x = -1, \ldots, 1$
`Theta_Delta2Npi`	2798	1.179, 1.090, 1.000, 1.090, 1.179
`ThetaDelta2NRad`	24	0.197, 0.598, 1.000, 0.598, 0.197
`DecayAngMEC`	1367	0.066, 0.533, 1.000, 0.533, 0.066
`VecFFCCQEshape`	4655	1.027, 1.014, 1.000, 1.014, 1.027

So the upstream weight production has already mirrored the morph knobs. The physical information lives entirely on $x \in [0, 1]$.

3.3 Spline vs Graph dials

GUNDAM builds each event's response from those knots according to `dialType`:

`Spline` A cubic spline through all knots. The default subtype is "not-a-knot" (same curve as ROOT's `TSpline3`); `dialSubType` can select `natural`, `catmull-rom`, or `monotonic`.

`Graph` Linear interpolation between neighbouring knots.

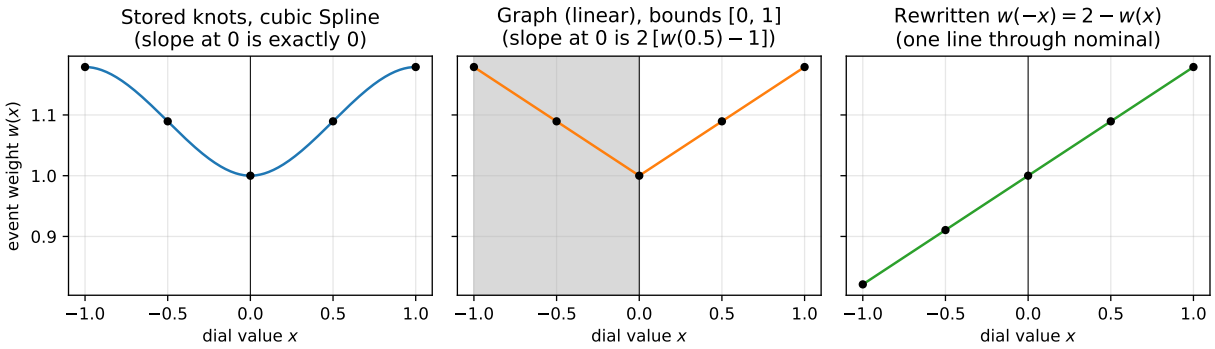


Figure 1: One `Theta_Delta2Npi` event from `gundam_v30.root`. Left: the default cubic spline through the symmetric knots has zero slope at the nominal point. Middle: a linear Graph dial restricted to $[0, 1]$ (grey region excluded by the bounds). Right: the knots rewritten as $w(-x) = 2 - w(x)$, giving one straight line through nominal.

For a symmetric set of knots, **any** smooth spline has zero slope at $x = 0$ (Fig. 1, left). The prediction μ_b is then flat to first order around nominal for every event, so:

- $\partial\mathcal{L}/\partial\theta = 0$ at $\theta = 0$ regardless of the data, so the nominal point is always a stationary point and Migrad has no gradient to follow;
- the data enters the curvature only through the $(1 - d_b/\mu_b)\,\partial^2\mu_b$ term of Eq. (5), which can be negative and can cancel the prior (Section 4).

This is what produced the post-fit `Theta_Delta2Npi` $= 0.00 \pm 3.7$ in the `emb` fake-data study.

3.4 Mirroring

GUNDAM's `useMirrorDial` with `mirrorLowEdge/mirrorHighEdge` folds the fit parameter back into the given range before the dial is evaluated (`DialInputBuffer.cpp`): a value past an edge reflects back, so with edges ± 1 a value of 1.5 is evaluated at 0.5 and 3.0 at -1. This only prevents extrapolation beyond the knots. Because the stored response is already symmetric, mirroring at $[-1, 1]$ or at $[0, 1]$ changes nothing, which is why setting `mirrorLowEdge: 0` had no effect.

3.5 Options and recommendation

1. **Graph dial, bounds $[0, 1]$, no mirroring** (current choice). Linear interpolation on $[0, 1]$ has slope $2[w(0.5) - 1] \neq 0$ at nominal, so the data supplies a gradient and the double minimum at $\pm\theta$ is removed. Costs: a kink in slope at the 0.5 knot, and nominal sits on the lower bound, so errors for a fit that ends near 0 are one-sided.

2. **Rewrite the negative knots as $w(-x) = 2 - w(x)$** in the input tree, then use Graph or Spline with bounds of about $[-3, 3]$. One straight line through nominal: no zero slope, no bound at nominal, normal Gaussian prior and Hesse behaviour. Negative values are a linear extrapolation with no model behind them; large negative pulls should be read as tension, not physics. Weights can go negative where $w(1)$ is large (e.g. `DecayAngMEC`); `minimumSplineResponse: 0` clamps them to 0, re-introducing a kink.

Mirroring ($w(-x) = w(x)$) is worse than either because it gives zero slope at nominal, a $\pm\theta$ degeneracy that a Gaussian prior at 0 cannot describe, and no way for the data to pull the knob "away from" the alternative model.

All four morph knobs should be treated the same way, even those that look well behaved in a particular fit: the zero slope is a property of the stored splines, not of the data. With the Graph dial, the 0.005 prior on `VecFFCCQEshape` is no longer needed.

Housekeeping. `CC1piTPi` and `LowQ2Suppression` are in `knoblist.py` and `GENIEDialsDef.yaml` but not in the knob lists of `makeParameterRootFile.C`; check that `gundaminput_geniesyst.root` actually contains priors for them.

Example dial definition:

```
- parameterName: "Theta_Delta2Npi"
  isEnabled: true
  dialSetDefinitions:
    - dialType: Graph
      minimumSplineResponse: 0
      applyOnDataSets: ["ICARUSNuMI_nueCCInclusive"]
      dialLeafName: "Theta_Delta2Npi"
      applyCondition: "[reco_is_data]==0"
```

4 Errors and the Hessian

4.1 From curvature to covariance

Near the minimum $\hat{\boldsymbol{\theta}}$ the likelihood is approximated by a quadratic,

$$\mathcal{L}(\boldsymbol{\theta}) \approx \mathcal{L}(\hat{\boldsymbol{\theta}}) + \tfrac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T H (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}), \qquad H_{ij} = \frac{\partial^2 \mathcal{L}}{\partial\theta_i \partial\theta_j}. \tag{3}$$

Since $\mathcal{L}$ is $-2\ln L$, the covariance is

$$C = 2\,H^{-1}, \tag{4}$$

and $\sigma_i = \sqrt{C_{ii}}$ is the Hesse error. For a single uncorrelated parameter this is the half-width at which $\mathcal{L}$ rises by 1.

4.2 What contributes to the curvature

Differentiating Eq. (2) twice for one parameter θ:

$$\frac{\partial^2 \mathcal{L}}{\partial \theta^2} = \underbrace{\frac{2}{\sigma^2_{\text{prior}}}}_{\text{penalty}} + 2\sum_b \Bigg[\underbrace{\frac{d_b}{\mu_b^2}\Big(\frac{\partial \mu_b}{\partial \theta}\Big)^2}_{\geq 0} + \underbrace{\Big(1 - \frac{d_b}{\mu_b}\Big)\frac{\partial^2 \mu_b}{\partial \theta^2}}_{\text{either sign}} \Bigg]. \tag{5}$$

Consequences:

- The first data term is never negative, so a dial with a real slope can only *tighten* the error below the prior.
- For an Asimov fit, $d_b = \mu_b$ at the minimum and the second data term vanishes; the post-fit error is then always $\leq$ the prior.
- A post-fit error **larger than the prior** is only possible when the second term is negative, i.e. the response is curved, the data disagrees with the prediction, and the slope is small. For `Theta_Delta2Npi`, $1/3.7^2 \approx 0.07$, so the data term cancelled about 93% of the prior curvature. That is a diagnostic, not a real 3.7σ uncertainty.

4.3 Marginal vs conditional errors

$\sqrt{C_{ii}}$ is the *marginal* error: it allows all other parameters to move. The *conditional* error, with all others fixed at the best fit, is $1/\sqrt{(C^{-1})_{ii}} \leq \sqrt{C_{ii}}$. They agree only when θ_i is uncorrelated with everything else. One-dimensional scans (Section 5) measure the conditional width.

4.4 When the Hesse error can be trusted

These are conditions for the symmetric Hesse error to be a valid *summary* of a parameter. Failing them does not mean the fit is bad; it means the parameter needs a different summary. The Hesse error assumes $\mathcal{L}$ is a parabola out to $\pm 1\sigma$. Check:

1. **Interval inside the bounds.** At a bound the sine transform has zero derivative and the external error is unreliable. More generally, if $\hat{\theta} \pm \sigma$ crosses a bound, the true interval is one-sided. For bounded parameters such as the morph knobs this is expected, not a failure; see Section 4.5.
2. **Not at a kink.** A Graph dial has a slope discontinuity at each knot (0.5 for the morph knobs); finite differences across it give a curvature that depends on which side is sampled.
3. **Smooth and parabolic over $\pm 1\sigma$.** Asymmetric or flat regions mean the quadratic approximation fails.

If any check fails, use a likelihood scan or MINOS (`errors: "Minos"`) and read the interval where $\Delta\mathcal{L} = 1$, clipped at the bounds.

4.5 Bounded parameters: the morph knobs on $[0, 1]$

With a Graph dial and bounds $[0, 1]$, a morph knob is a different statistical object from a multisigma knob, and the Gaussian-error language above does not apply to it directly.

What the prior is. The Graph is the dial *response*; the prior is the penalty term θ^2/σ^2 with prior 0 and $\sigma = 1$. Truncated by the bounds, it is a half-Gaussian on $[0, 1]$: staying at the nominal model costs nothing, and switching fully to the alternative model costs $\Delta\text{llh} = 1$. This encodes the usual convention that the alternative model is a "1σ" variation. A different belief needs a different prior: for example, if both models are considered equally plausible, a flat prior on $[0, 1]$ (no penalty) is more appropriate.

The prior width already spans the whole range. Since $\sigma = 1$ equals the width of the allowed range, $\hat{\theta} \pm \sigma$ will cross a bound for any morph knob the data does not strongly constrain. That is a consequence of the parametrization, not a symptom of a bad fit.

Sitting at a bound can be the right answer. $\hat{\theta} = 0$ means the data prefers the nominal model; $\hat{\theta} = 1$ means it prefers the full alternative. Both are physically sensible outcomes for an on/off effect. What to check instead:

- In a scan, `llh` should *rise* moving away from the bound into the allowed region. If it falls, the minimizer stopped early.
- Report a one-sided interval from the scan, e.g. $\theta < \theta_1$ where $\Delta\text{llh}(\theta_1) = 1$ relative to the value at the bound, rather than the Hesse $\pm$ error.
- For an interior best fit, read the two crossings of $\Delta\text{llh} = 1$ separately, clipped at 0 and 1; they will generally be asymmetric.

Where it does matter: throws. The toy generator and `gundamCalcXsec` throw parameters from the post-fit (Hesse) covariance as a multivariate Gaussian. In GUNDAM 1.9 a throw that puts any parameter outside its bounds is rejected and the whole vector re-thrown (`ParametersManager.cpp`), so morph knobs are sampled from a truncated Gaussian whose width comes from the Hesse matrix. Two consequences:

- If the Hesse width is unreliable (best fit at or near a bound), the throws inherit that error. Compare the spread of thrown morph values with the scan.
- The acceptance is low: a knob at 0 with width 1 lands in $[0, 1]$ about 34% of the time, so four such knobs accept roughly 1% of throws.

If this becomes a problem, rewriting the knots as $w(-x) = 2 - w(x)$ and removing the bounds (Section 3) makes the morph knobs ordinary Gaussian parameters, for which Hesse and Gaussian throws are both valid. That is the main practical argument for that option.

5 Likelihood scans

5.1 What the scan produces

With `enablePostFitScan: true` (and/or `enablePreFitScan`) in `fitterEngineConfig`, GUNDAM steps each free fit parameter across a range around its current value, holding all others fixed, and records at each point:

`llhPenalty` the prior term of Eq. (2). Independent of the data. For an uncorrelated knob it is the parabola $(\theta - \theta^{\text{prior}})^2/\sigma^2$; for correlated knobs its minimum can be shifted; for the template parameters it is flat at 0.

`llhStat` the Poisson data term. Shows what the data alone says about the knob.

`llh` the sum, i.e. what the fit minimizes.

Optional extras (`llhStatPerSample`, per-bin and per-sample weight scans) are enabled in `scanConfig`.

GUNDAM 1.9 bug. With the post-fit scan enabled, 1.9 crashed with `TDirectoryFile::mkdir: An object with name Template_Parameter_true_leading_electron_energy exists already`, followed by a segfault. The scanner writes `<full title>_CurrentPar`, whose name contains a `/`; the write helper turns it into an unchecked `mkdir`, which fails on the second parameter of a set and returns a null directory. Fixed locally in `ParameterScanner.cpp` by replacing `/` with `_` in that name, as is already done for the scan graphs. Requires a rebuild of 1.9.

5.2 Reading the shapes

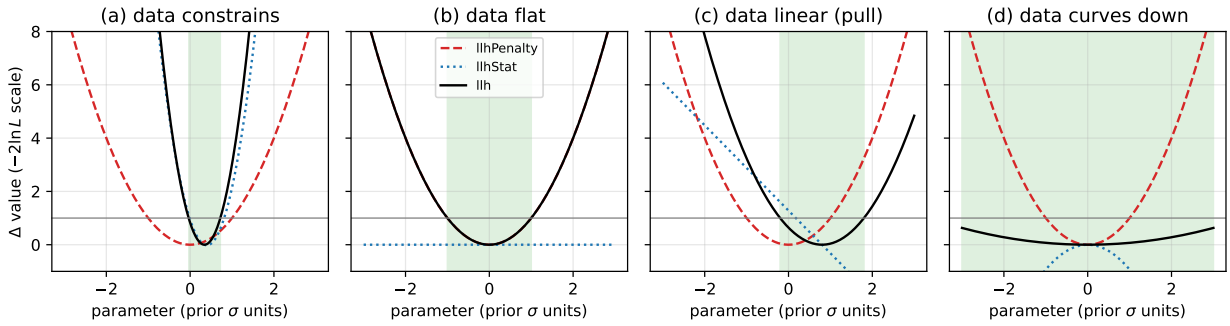


Figure 2: Illustrative scan shapes (toy curves, not from a fit), in units of the prior σ with prior at 0. The shaded band is where $\Delta\text{llh} \leq 1$. (a) Data constrains the knob: narrow total, error well below 1. (b) Data flat: total equals the prior, error ≈ 1. (c) Data linear: total is the prior parabola shifted by half the slope; error still ≈ 1. (d) Data curves downward: total is much flatter than the prior, error $\gg 1$, as for `Theta_Delta2Npi` with the original spline.

llhStat shape	Interpretation
Deep, narrow valley	Data constrains the knob; post-fit error well below prior.
Flat	Knob does not change the selected-sample prediction; error $\approx$ prior.
Linear	Data pulls one way with no curvature; best fit shifts by about half the slope (in χ^2 per σ), error $\approx$ prior.
Curving downward	Data term reduces the curvature; error larger than prior. Check the dial shape and whether the fit sits at a stationary point.
Minimum far from prior	Data and prior disagree. In a fake-data study, often the knob absorbing the injected variation; otherwise, a mismodelling to understand.
Kinks, steps, jagged	Dial problem: interpolation knots, weights clamped by `minimumSplineResponse`, or bad splines.

5.3 Using scans to decide whether the fit succeeded

1. **Minimum at the centre.** Post-fit scans are centred on the best fit, so `llh` should have its minimum there. A minimum elsewhere means the minimizer did not converge in that direction.

2. **Width matches Hesse.** Read where `llh` rises by 1 on each side. If it is close to $\pm\sigma_{\text{Hesse}}$ and parabolic, the Hesse error is fine. If it is asymmetric, kinked, or hits a bound first, quote the scan interval instead. A scan narrower than Hesse is expected for correlated knobs (conditional vs marginal); check the post-fit correlation matrix.

3. **What sets the error.** Compare the steepness of `llhStat` and `llhPenalty`. Data-dominated knobs should have post-fit errors well below 1; prior-dominated knobs about 1; anything above 1 needs explaining.

4. **Physical behaviour.** Smooth curves indicate healthy dials; kinks or steps point at the dial definition or input splines.

5.4 Unconstrained systematics are not bad systematics

A flat or linear `llhStat` means the fit cannot reduce that uncertainty, so its full prior propagates into the result. That is correct behaviour. Such knobs must be kept: they often act through the efficiency and background subtraction in the extracted cross section, which the selected sample may not be sensitive to. Dropping a knob is only justified if it has no effect on anything reported: check the $\pm1\sigma$ event-rate variations (`preFit/oneSigma`, `preFit/varyEventRates`) and the change in the `gundamCalcXsec` throw spread when the knob is fixed.

6 Why the extracted cross-section uncertainty is large

The post-fit *distribution* (reconstructed event rate) carries a small uncertainty, a few percent per bin, while the extracted *cross section* in true-E_e bins is 20–32%. Both come from the same fit and the same post-fit covariance. This section explains the gap, using a breakdown study driven by `run_frac_uncertainty.sh`, which re-runs the fit, `gundamCalcXsec`, and the toy generator with

individual systematic groups switched on or off (variants `template_only`, `xsec_only`, `flux_only`, `detector_only`, `all`, and the N−1 variants `no_xsec`/`no_flux`/`no_detector`). All numbers below are Asimov, at 2.5×10^{20} POT.

6.1 The flux–template degeneracy

The predicted events in true bin i factorize as

$$N_i \;=\; \Phi \cdot \sigma_i \cdot \epsilon_i \cdot T \cdot (\dots), \tag{6}$$

with flux Φ, cross section σ_i, efficiency ϵ_i, and number of targets T. The template parameters t_i are *free* (no prior covariance; bounds $[0, 3]$ only), one per true-E_e bin, and multiply σ_i. Every factor in Eq. (6) enters multiplicatively, so a shift in Φ, ϵ_i, or any reweighting systematic produces the same change in N_i as a change in t_i. With ten free templates spanning all per-bin rescalings, any systematic whose effect is a rescaling of the true-E_e spectrum is reproduced exactly by the templates.

The fit therefore constrains only the *product* $t_i \times (\text{syst})$, which the data pins down, not the split. Along the degenerate direction the data term of Eq. (2) is flat; the tie-break is the penalty. A systematic has a Gaussian prior ($\sigma = 1$) and stays at it; the template has no prior and absorbs the shift, its post-fit error inflating. "The template absorbs the systematic" and "the systematic stays at its prior" are the same statement. This extends Section 5 ("Unconstrained systematics are not bad systematics") to the normalization of the signal itself.

A mechanical picture. (Figs. 3 and 4) Think of each parameter as a bead the fit displaces from nominal. The prior attaches a spring of stiffness $2/\sigma^2_{\text{prior}}$ (the penalty term of Eq. (5)); the data attaches a second spring whose stiffness is the first data term, $2\sum_b (d_b/\mu_b^2)(\partial\mu_b/\partial\theta)^2$, and therefore grows with the event counts d_b. The template parameters carry *no* prior spring. Pushed along a degenerate direction, the free templates—restrained by nothing—slide all the way to absorb it, while a systematic, held by its prior spring, barely moves. Low event counts make the data spring weak, so even a systematic with some distinguishing power stays pinned near its prior: this is why, at this sample size, the systematics scarcely move and the templates take up essentially all of the variation. More statistics stiffens the data spring, but only along the directions the response matrix makes distinguishable (the diagonal test below); along an exactly degenerate direction the data spring has zero stiffness no matter the counts, which is the $\sigma = 1.000$ limit.

Reading the post-fit covariance in normalized fit space confirms it: in the `all` fit the flux-beamline and detector sets sit at $\sigma \approx 0.998$ and 0.991 (prior = 1), essentially unconstrained, while the templates inflate to $\sigma \approx 1.6$–2.1.

6.2 Why the distribution is precise but the cross section is not

The distribution is the *product* the data constrains; the cross section is *one factor* of that product. Isolating the factor means dividing out the systematic, which is known only to its prior:

$$\sigma_i \;\propto\; \frac{N_i}{\Phi\,\epsilon_i} \quad\Longrightarrow\quad \left(\frac{\delta\sigma_i}{\sigma_i}\right)^2 \approx \underbrace{\left(\frac{\delta N_i}{N_i}\right)^2_{\text{stat}}}_{\text{distribution}} + \left(\frac{\delta\Phi}{\Phi}\right)^2_{\text{prior}} + \left(\frac{\delta\epsilon_i}{\epsilon_i}\right)^2_{\text{prior}} + \dots \tag{7}$$

The distribution carries only the first term; the cross section carries all of them. By analogy (Fig. 5): measuring the area of a rectangle to 1% says little about its length if the width is known

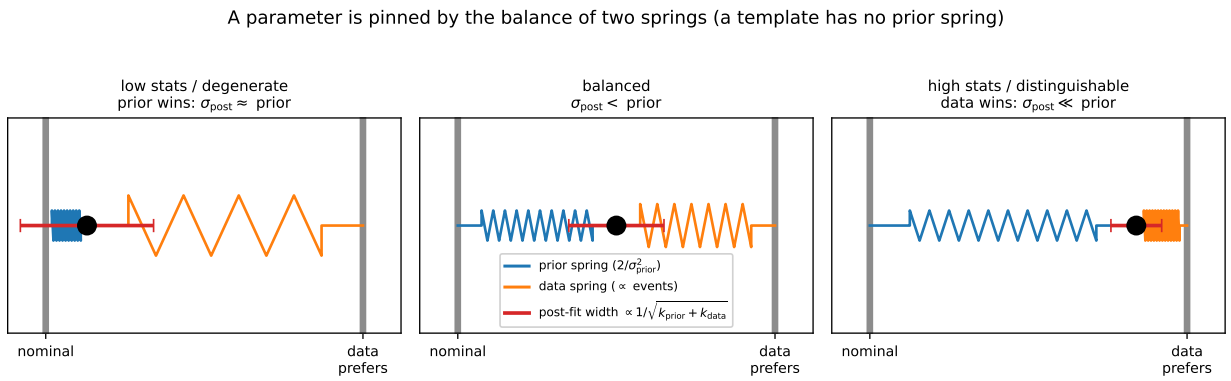


Figure 3: How well a single parameter is pinned is the balance of two springs (Eq. (5)): a prior spring of stiffness $2/\sigma^2_{\text{prior}}$ tying it to nominal, and a data spring whose stiffness grows with the event count, pulling it toward the data-preferred value. The post-fit width is $\propto 1/\sqrt{k_{\text{prior}} + k_{\text{data}}}$ (red bar). Left: with few events, or along a degenerate direction, the data spring is limp, the prior wins, and the parameter stays near nominal with $\sigma_{\text{post}} \approx$ prior. Right: with many events and a distinguishable response the data spring is stiff and the parameter is pulled to the data value and tightly constrained. A template parameter has no prior spring at all (left anchor removed); it is held only by the data spring, and in an exactly degenerate direction by nothing.

only to 15%—the length (cross section) inherits the width's (systematic's) prior, not the area's (distribution's) precision.

In the same true-E_e bins (diagonal binning, `all`):

Quantity (same bins, same fit)	mean fractional unc.
Post-fit distribution	8.6%
Extracted cross section	19.1%

6.3 Pre-fit vs post-fit statistical band: why the post-fit is larger

In an Asimov study the *statistical* band of the reconstructed distribution comes out larger *after* the fit than before, which is surprising until the free templates are accounted for. For the stat-only setup (`template_only`, no systematics), with the toy generator throwing `enableStatThrowInToys` and `enableEventMcThrow` on top of the parameters:

Quantity (`template_only`, 10 bins)	mean frac. unc.
Distribution, pre-fit band	2.9%
Distribution, post-fit band	5.8%
Template post-fit Hesse error (true bins)	9.0%

The only difference between the two bands is the parameter throw:

- **Pre-fit** (`--use-prefit`) throws parameters from the *prior* covariance. The templates are free—no prior covariance, only `parameterLimits`—so there is nothing to throw for them (a flat prior cannot be sampled; GUNDAM leaves them at nominal). The pre-fit band is then only the stat and MC throws: the Poisson/MC scatter of a *fixed* nominal prediction, with no signal-normalization uncertainty in it at all.

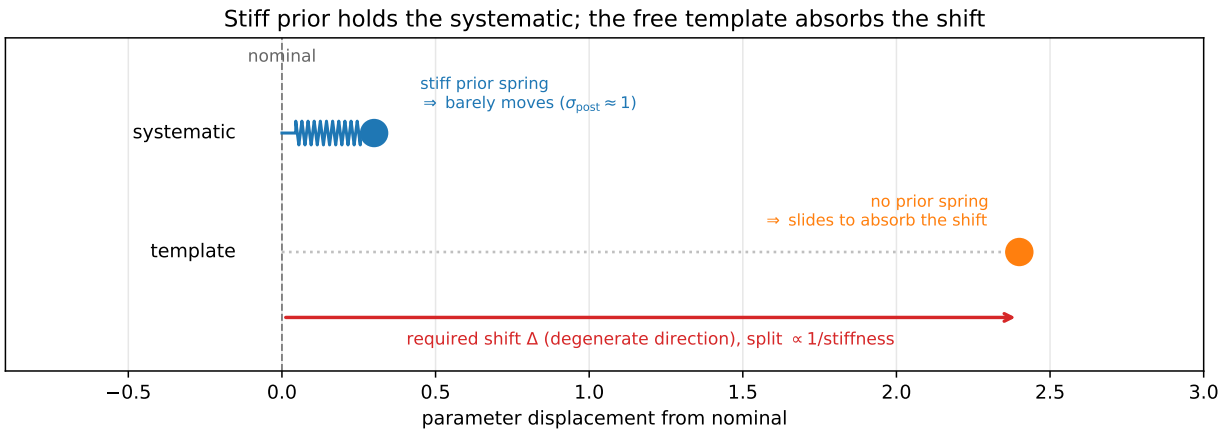


Figure 4: Spring picture of the degeneracy. The prior attaches each systematic to its nominal value with a stiff spring (stiffness $2/\sigma^2_{\text{prior}}$, plus a data spring that grows with the event count); the free template parameter has no prior spring. A shift along the degenerate direction is shared between the beads in inverse proportion to their stiffness, so the free template slides to absorb it while the systematic stays near its prior. Low event counts weaken the data spring, pinning the systematic even harder.

- **Post-fit** (`--use-bf`) throws from the post-fit Hesse covariance, which now carries the template errors ($\sim$9% per true bin) that the data created by constraining the free signal strength. Throwing these adds in quadrature: $\sqrt{5.8^2 - 2.9^2} \approx 5.0\%$ is the template contribution projected into reco space, smaller than the 9% true-bin error because the response matrix averages it down.

This is the "free templates have no prior spring" point (Fig. 3) applied to the toy bands. For a parameter *with* a prior (a systematic), the Asimov pre- and post-fit bands are nearly equal, because the fit barely moves it ($\sigma \approx$ prior before and after). The templates are the exception: their uncertainty is undefined before the fit and is *created* by it. So the pre-fit distribution band is not a complete statistical uncertainty for a free-template analysis—it omits the signal normalization entirely—and the post-fit band is the one to quote.

6.4 Decomposition of the total uncertainty

The measurement is *not* statistics-limited. The fiducial signal yield is ~ 450 events per true-E_e bin (~ 4540 total), a raw weighted-Poisson uncertainty of only $\sim 2.7\%$ per bin. The 23.6% total (`all`, nominal reco binning) is a quadrature sum of several comparable contributions, isolated as $\sqrt{(\text{variant})^2 - (\texttt{template_only})^2}$:

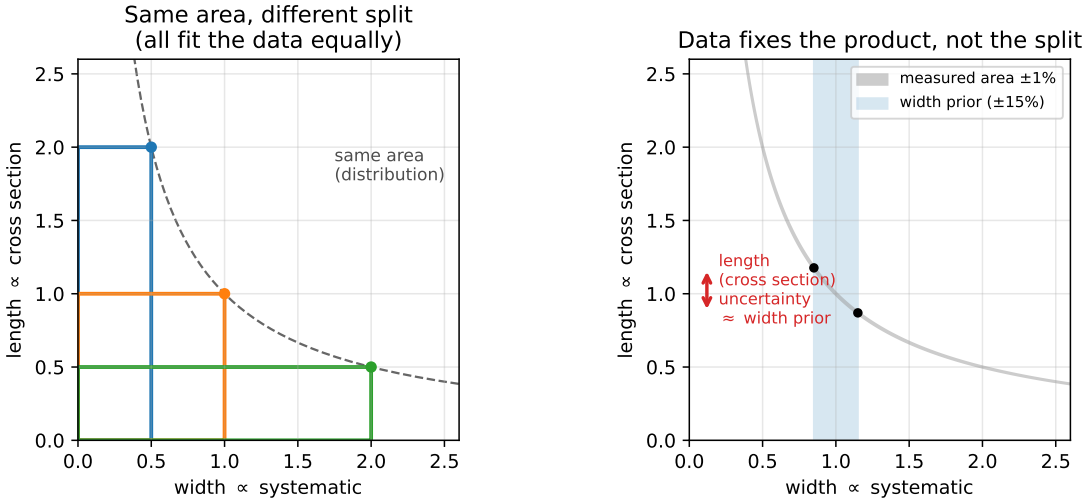


Figure 5: Rectangle picture. Left: rectangles of equal area have their corners on the hyperbola $\ell = A/w$; the data fixes the area (the distribution) but not the split into length ($\propto$ cross section) and width ($\propto$ systematic). Right: the area is measured tightly (a $\pm 1\%$ band), but a wide prior on the width ($\pm 15\%$) leaves the length uncertain by about the width's prior, not the area's precision—the distribution-versus-cross-section gap.

Contribution	frac. unc.	share of variance
GENIE	12.2%	27%
Detector	11.6%	24%
Background-diluted fit statistics	10.0%	18%
Flux normalization	9.1%	15%
Flux-beamline	8.2%	12%
Raw signal Poisson	2.7%	1%
Total (quadrature)	23.6%	—

Two contributions are flux-related but distinct: the *flux normalization* (the `Constant flux` entry in `fitSamples_TrueEleE.yaml` `normaliseParameterList`, $\sigma = 9.13\%$ = hadron production $8.99\% \oplus$ beam focusing 1.62%, applied by `gundamCalcXsec` to every bin) and the *flux-beamline* parameter set. No single term dominates, so no one systematic is a decisive lever. Zeroing the flux normalization (`run_frac_uncertainty.sh --zero-flux-norm`) moves the total only from 23.6% to 21.3%, because a 9% term inside a 24% total is only $\sim$15% of the variance.

6.5 How each contribution is computed (`--decompose`)

The individual lines are isolated by re-running `gundamCalcXsec` on a single fit and toggling one throw at a time, then taking quadrature differences. The driver is `run_frac_uncertainty.sh --decompose`.

The two "flux" contributions are distinct.

- **Flux norm** — the overall flux *normalisation* (Φ in $N/(\Phi\, T\, \epsilon)$), the `Constant flux` entry of

the `normaliseParameterList` in `fitSamples_TrueEleE.yaml` ($\sigma = 9.13\%$, 100% correlated across bins, applied by `gundamCalcXsec` to the cross-section definition — not a fit parameter).

- **Flux beamline** — the NuMI beamline/focusing *systematic* (energy-dependent flux shape), the fit parameter set `parameterSet_Beamline.yaml` ("Flux Systematics Beamline").

The stat-only decomposition. The cross-section error from `gundamCalcXsec` is the spread of the $N = 1000$ toys; each toy throws some subset of: the post-fit parameter covariance, a Poisson data fluctuation (`enableStatThrowInToys`), an MC-statistical fluctuation (`enableEventMcThrow`), and the normaliseParameters. Running on the `template_only` fit with one toggle at a time isolates each piece:

Line (variant dir)	statThrow	mcThrow	normalisers kept
Template (`template_base`)	off	off	none
Stat (`statthrow`)	**on**	off	none
Efficiency stat (`mcthrow`)	off	**on**	none
Flux norm (`fluxnorm`)	off	off	`Constant flux`
Targets (`targets`)	off	off	`Number of targets`

Each contribution is the quadrature difference from the template base $t \equiv$ `template_base`,

$$C_X = \sqrt{\max\left(0,\ f_X^2 - f_t^2\right)}, \qquad f_X = \frac{\text{bin error}}{\text{bin value}} \times 100, \tag{8}$$

read per bin from `calcXsec/histograms/TrueEleE_XS_TH1D`. *Template* is f_t itself: the template parameters' post-fit (Hesse) covariance propagated to the cross section — the fit's statistical constraint on the signal, which carries the migration structure (hence it rises at low energy; see the diagonal-response test below). The systematic sets (GENIE, flux-beamline, detector) are isolated with the same Eq. (8) but above the *full* `template_only` (all throws and normalisers on), so every line adds in quadrature to Total.

Caveat. The efficiency-stat line is only the *signal-template* MC noise. Each systematic's own dial-reweighting MC noise stays inside that systematic's line; to split a set's physical part from its MC noise, re-run it with `--no-mc-throw`.

6.6 The diagonal-response (perfect-resolution) test

To separate the resolution penalty from the irreducible degeneracy, `run_frac_uncertainty.sh --diagonal` bins the fit sample on `true_leading_electron_energy` (`binning_TrueEleE.txt`), so the "reco" variable equals the true variable and the response matrix is the identity.

- The total falls only from 23.6% to 19.1%, and the stat-only (`template_only`) term from 13.6% to 11.3%: a 15–20% relative improvement. Resolution is *not* the dominant driver.
- The flux-beamline and detector sets go to $\sigma = 1.000$ *exactly*, more degenerate than in reco space. Binning on true energy removes the migration dimension, so every systematic acts purely as a per-bin rescaling—indistinguishable from the templates.

This inverts a natural intuition: resolution/migration is the *only* feature that gives a systematic a fingerprint distinct from a per-bin template rescaling, so it is the mechanism by which the fit constrains systematics. Perfect resolution is the *best* case for constraining them; the diagonal limit (no migration) is the worst, and makes the degeneracy exact. The gap of the post-fit σ below 1 in reco space (0.991, 0.998) is exactly the small part the response matrix makes distinguishable.

With a diagonal response the free templates are exactly degenerate with any reweighting systematic, so the N−1 variants are ill-posed and Migrad diverges to NaN (the recombination dial is where the NaN first surfaces); the driver script skips the N−1 variants in diagonal mode and still produces the individual-contribution plots.

6.7 More statistics does not help

Only the statistical term scales with POT; the systematic floor is fixed priors. Projecting the diagonal, flux-norm-removed budget with stat $\propto 1/\sqrt{\text{POT}}$:

POT	stat term	total xsec unc.
2.5×10^{20} (now)	6.1%	15.9%
1.0×10^{21}	3.1%	14.9%
2.5×10^{21}	1.9%	14.7%
infinite	0.0%	14.6%

Even infinite statistics reaches only $\sim$14.6%: the measurement is systematics-limited. In reco space, more statistics marginally tightens the small non-degenerate part of each systematic ($\sigma < 1$); in the diagonal limit ($\sigma = 1$) it does nothing.

6.8 Summary: three mechanisms that inflate the cross section

The post-fit distribution uncertainty ($\sim$ 6% per bin) and the extracted cross-section uncertainty ($\sim$20–32% per true bin) come out of the same fit and the same post-fit covariance. Three distinct mechanisms account for the gap:

1. **Un-smearing (resolution / migration).** The reconstructed distribution is the true-bin templates *smeared down* by the response matrix — each reco bin averages over several true bins. The cross section un-smears back to true bins, amplifying every per-bin uncertainty by $\sim 2\times$ (stat 5.8% $\rightarrow$ 9.0%, and a comparable factor on each systematic). Removable only by better resolution; the diagonal-response test sets the floor. It is largest at threshold, where the resolution ($\sim$46%) is far wider than the true-bin width.

2. **Dividing by flux and efficiency.** The cross section is $N/(\Phi\, T\, \epsilon)$. The flux normalisation (9.13%) is baked into the definition and absent from the distribution. The detector and flux-beamline dials act on the selected sample but *not* the truth denominator, so they do not cancel in the efficiency ϵ and survive as a cross-section uncertainty *even at perfect resolution* ($\sim$6% for detector in the diagonal test).

3. **Loss of template shielding for the systematic priors.** In the distribution the free templates absorb the unconstrained systematics to keep the prediction pinned to the data, so GENIE, detector and flux-beamline are nearly invisible there ($\lesssim 1\%$ for detector). The cross section *is* the template, so their full priors re-appear: GENIE $\sim$12%, detector $\sim$12%, flux-beamline $\sim$8%. This is the dominant bucket, removable only by an external constraint that breaks the degeneracy.

These are mechanisms, not a strict additive partition: detector and flux reach the cross section through more than one (mechanism 1 un-smears them, mechanism 2 carries them through the efficiency). Quantitatively, in the lowest-energy bin the un-smearing is the single largest piece ($\sim 27\%$ of the 32.9%), while integrated over the spectrum the systematic priors (mechanism 3) dominate.

6.9 What would reduce it

1. **Better E_e resolution.** Shrinks the off-diagonal response and gives systematics a larger distinguishable fingerprint, letting the fit pull them below their priors. Modest, as the diagonal test bounds the gain.

2. **An external constraint.** A high-statistics control or sideband sample that measures flux and detector directly, so they reach the signal fit already narrower than their priors—breaking the template degeneracy before it reaches the cross section. This is the only route to the few-percent level.

3. Statistics in the signal sample alone, and removing the flux normalization, are *not* effective routes on their own.

7 Checklist for a fit

1. Migrad converged (status 0, small EDM); Hesse matrix accurate and positive definite.

2. Parameters at a bound are either physically sensible (e.g. a morph knob choosing one model) or understood; for these, `llh` rises into the allowed region and the uncertainty is quoted as a one-sided interval from a scan, not the Hesse error.

3. No post-fit error exceeds its prior without an explanation.

4. Post-fit scans have their minimum at the centre, are smooth, and have $\Delta\text{llh} = 1$ widths consistent with Hesse (allowing for correlations).

5. Morph knobs use a dial with non-zero slope at nominal (Graph on $[0, 1]$, or rewritten knots).

6. In fake-data studies, the knobs that move are the ones expected to absorb the injected variation.